ESM_1 Supplementary analyses

Online Resource 1 for the manuscript on employment stress and career choice among Chinese college students

This document reports two sets of supplementary analyses that the main text refers to but does not report in full. Section 1 gives the test used to decide whether the two survey waves could be analysed as one pooled sample (Table S1, Fig. S1). Section 2 gives the comparison between the main sample and the pooled sample for the descriptive profile addressed in research question 1 (Figs. S2–S6). All estimates are reported exactly as produced by the analysis scripts.

1. Pooling eligibility of the two survey waves

1.1 Design and test statistics

The main sample (N = 1,026; collected May–June) and the second wave (N = 553; collected in September) differ in composition and in the time at which they were collected, so their eligibility for pooling was tested rather than assumed. Let X denote the stress-source index, standardised on the pooled sample (SD = 1), and B the wave indicator (0 = main sample, 1 = second wave). Three nested models were estimated for each outcome:

$M0$: $Y \sim X$ + sex + grade + major + channel
$M1$: $M0 + B$
$M2$: $M1 + B \times X$

$M1$ adds a wave-specific intercept difference (δ), and $M2$ a wave-specific slope difference (θ). The step from $M1$ to $M2$ is the decisive one: θ = 0 means that the stress–choice association has the same strength in both waves, so that pooling averages one association; θ ≠ 0 means that it averages two. For continuous outcomes the nested comparison is an F test, $F = [(\mathrm{RSS}_r - \mathrm{RSS}_u)/q] / [\mathrm{RSS}_u/(n - k_{M2})]$. Because homoscedasticity does not hold in these data (Breusch–Pagan p = .0003 for health impact and p = .0067 for the expectation-impact index), a heteroskedasticity-robust Wald test with HC3 standard errors was computed alongside and is treated as the primary criterion. For the nominal outcome (four-category intention) the comparison is a likelihood-ratio test, $\mathrm{LR} = -2(\log L_r - \log L_u) \sim \chi^2(q)$. The five θ tests were corrected for multiple testing with the Benjamini–Hochberg procedure. Majors with fewer than 50 cases in the pooled sample (medicine, agriculture, law, education, arts and sport) were merged into a residual category.

1.2 Results

Neither the intercept difference nor the slope difference reached significance for any outcome once the five tests were corrected (Table S1, Panel A). Four of the five θ estimates lie close to zero with confidence intervals centred on it. The exception is the expectation-impact index (Q11): θ = −0.060, 95% CI [−0.114, −0.007], uncorrected p = .028, Benjamini–Hochberg p = .138 (the same value under Bonferroni). Under the correction adopted for the five tests homogeneity is therefore not rejected, but this outcome is the weakest link in the table and is reported as such throughout.

The wave-specific slopes in Panel B show where the estimate comes from. All four slopes carry the same sign, and the difference is one of magnitude only: for the expectation-impact index the second-wave slope (+0.065) is roughly half the main-sample slope (+0.131), whereas for health impact the two slopes are close to each other (+0.342 and +0.363). Two sensitivity analyses are reported in Panel C. Excluding respondents whose Q9 selections exceeded the permitted number (98 in the main sample and 61 in the second wave; N = 1,420), the same outcome remains borderline, θ = −0.056, 95% CI [−0.111, −0.001], p = .046, and the intercept difference becomes significant (p = .038); the instability of this outcome therefore reproduces under both specifications rather than being an artefact of one. In the postgraduate-engineering subgroup (N = 179; 20 and 159), adjustment for sex and channel removes the apparent gap, θ = +0.039 (p = .822) and δ = −0.343 (p = .155), indicating that the unadjusted difference in this subgroup reflected compositional confounding rather than the wave itself.

On this basis the analyses reported for research question 3 use the pooled sample (N = 1,579), with the wave indicator retained in every model, and the interacted estimate for the expectation-impact index is reported next to the pooled estimate. The descriptive profile of research question 1 is not pooled; the reason is set out in Section 2.

Table S1 Pooling eligibility of the two survey waves. δ is the wave-specific intercept difference (M1 versus M0) and θ the wave-by-stress-source interaction (M2 versus M1); *X* is the stress-source index standardised on the pooled sample (*SD* = 1). Panel A reports the nested-model comparison for the five outcomes, with 95% confidence intervals in brackets, HC3-robust *p* values for θ, and Benjamini–Hochberg-adjusted *p* values across the five θ tests. The four-category intention outcome is compared by likelihood ratio (χ^2 with 3 *df*) rather than by a coefficient. Panel B reports the slope of *X* estimated separately within each wave and in the pooled sample. Panel C reports the two sensitivity analyses. All models control for sex, grade, major and collection channel.

Panel A. Nested-model comparison

Outcome (question)	N (main / second)	δ [95% CI]	*p*(δ)	θ [95% CI]	*p*(θ), HC3	*p*(θ), BH
Public-sector preference (Q10)	1,579 (1,026 / 553)	+0.027 [−0.165, 0.219]	.786	−0.039 [−0.158, 0.081]	.527	.773
Flexible / new-form preference (Q10)	1,579 (1,026 / 553)	−0.008 [−0.170, 0.154]	.926	−0.025 [−0.127, 0.078]	.639	.773
Expectation-impact index (Q11)	1,554 (1,013 / 541)	+0.081 [−0.026, 0.187]	.138	−0.060 [−0.114, −0.007]	.028	.138
Health impact (Q5)	1,579 (1,026 / 553)	−0.117 [−0.293, 0.058]	.191	+0.016 [−0.090, 0.122]	.773	.773
Four-category intention (Q7)	1,579 (reconstructed)	LR = 2.30 (3 *df*)	.512	LR = 3.82 (3 *df*)	.282	.705

Panel B. Slope of the stress-source index within each wave

Outcome (question)	Main sample (*N* = 1,026)	Second wave (*N* = 553)	Pooled (*N* = 1,579)
Public-sector preference (Q10)	+0.143	+0.093	+0.143
Flexible / new-form preference (Q10)	+0.070	+0.040	+0.067
Expectation-impact index (Q11)	+0.131	+0.065	+0.132
Health impact (Q5)	+0.342	+0.363	+0.343

Panel C. Sensitivity analyses

C1. Excluding respondents whose Q9 selections exceeded the permitted number (98 main, 61 second wave; *N* = 1,420)

Outcome (question)	*p*(δ)	θ [95% CI]	*p*(θ), HC3
Public-sector preference (Q10)	.489	−0.034 [−0.156, 0.089]	.593
Flexible / new-form preference (Q10)	.830	−0.044 [−0.151, 0.063]	.420
Expectation-impact index (Q11)	.038	−0.056 [−0.111, −0.001]	.046
Health impact (Q5)	.334	+0.015 [−0.097, 0.128]	.790

C2. Postgraduate-engineering subgroup, health impact (*N* = 179; 20 main, 159 second wave), adjusted for sex and collection channel: δ = −0.343 (*p* = .155), θ = +0.039 (*p* = .822).

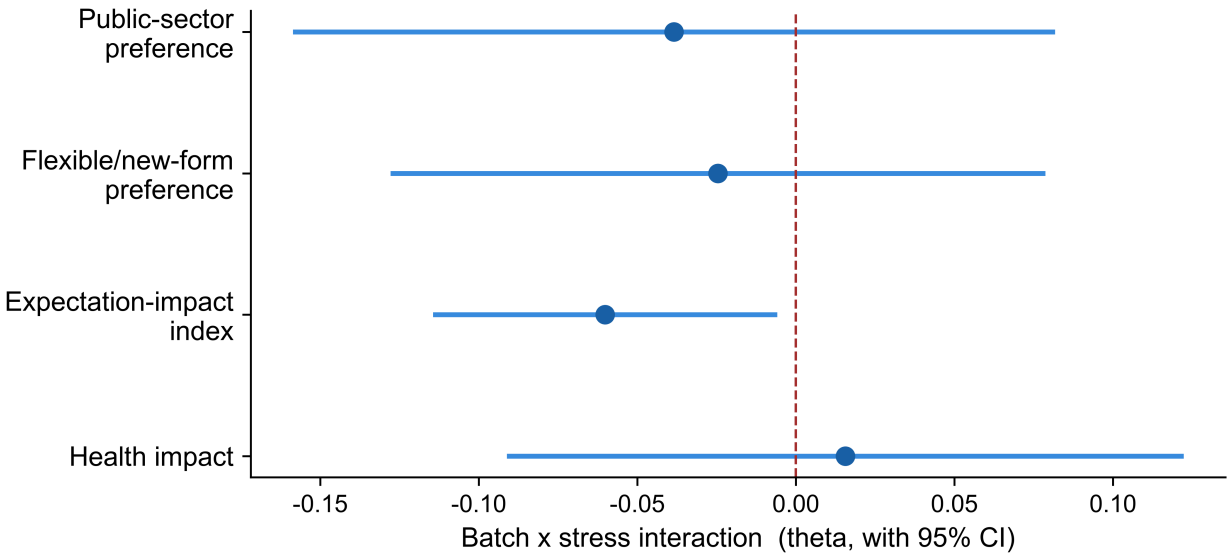


Fig. S1 Wave-by-stress interaction (θ) for the four continuous outcomes, with 95% confidence intervals. Filled circles are the point estimates of the interaction term in model M2; horizontal lines are HC3 95% confidence intervals; the dashed vertical line marks θ = 0, that is, the same stress–choice association in both waves. Every interval crosses zero once the five tests are corrected (Table S1, Panel A). Rows are ordered by the point estimate. The four-category intention outcome is tested by likelihood ratio and therefore appears in Table S1 only.

2. Sensitivity of the descriptive profile to pooling

2.1 Design

The question here differs from that in Section 1. A pooled regression can absorb a wave main effect through an indicator and can test slope homogeneity through an interaction term; a descriptive profile has neither device. Whether the profile should be computed on the pooled sample is therefore a question about the sample rather than about a coefficient. The same functions were applied to the main sample (N = 1,026) and to the pooled sample (N = 1,579), and the two outputs were compared item by item on three criteria: whether the ordering of items is preserved (Spearman ρ and top-three overlap), whether the level of each indicator is preserved (Cohen's d), and whether the composition of the sample is preserved (percentage-point shift per category).

2.2 Results

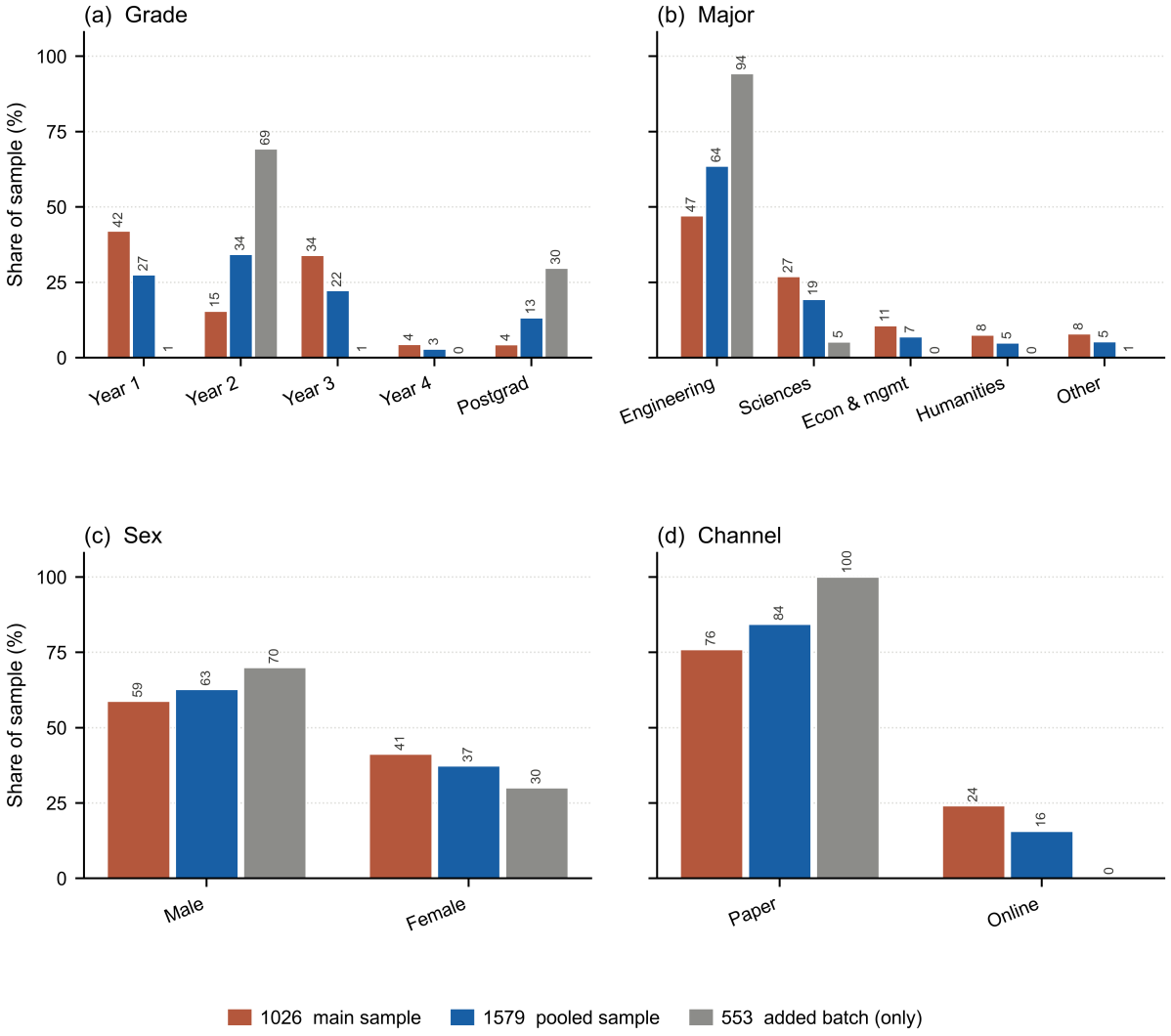


Fig. S2 Composition of the main sample (N = 1,026; dark red), the pooled sample (N = 1,579; blue) and the additional second wave (N = 553; grey), by (a) year of study, (b) major, (c) sex and (d) collection channel. Bars give the share of each sample falling in a category, with the value printed above each bar. The three samples are shown side by side rather than stacked so that the category labels stay on the axis and one colour refers to one sample throughout.

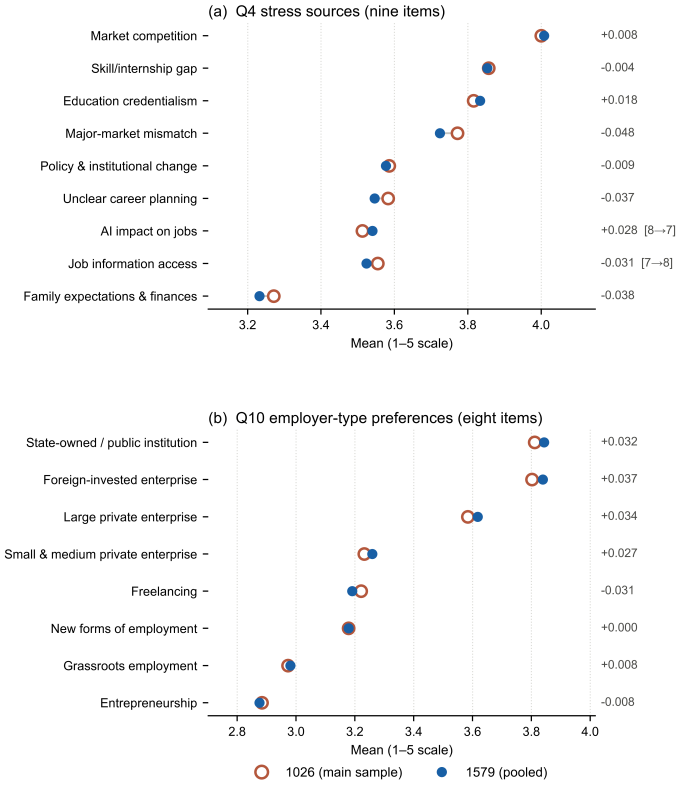


Fig. S3 Item-level means in the main sample (open circles) and the pooled sample (filled circles) for (a) the nine stress sources and (b) the eight employer preferences. Rows are ordered by the pooled mean. The annotation on the right of each row gives the difference in means and, where the rank changes, the change in rank (main sample → pooled sample). Horizontal position, not bar length, carries the comparison. Spearman's ρ between the two rankings is 0.983 for the nine stress sources and 1.000 for the eight preferences.

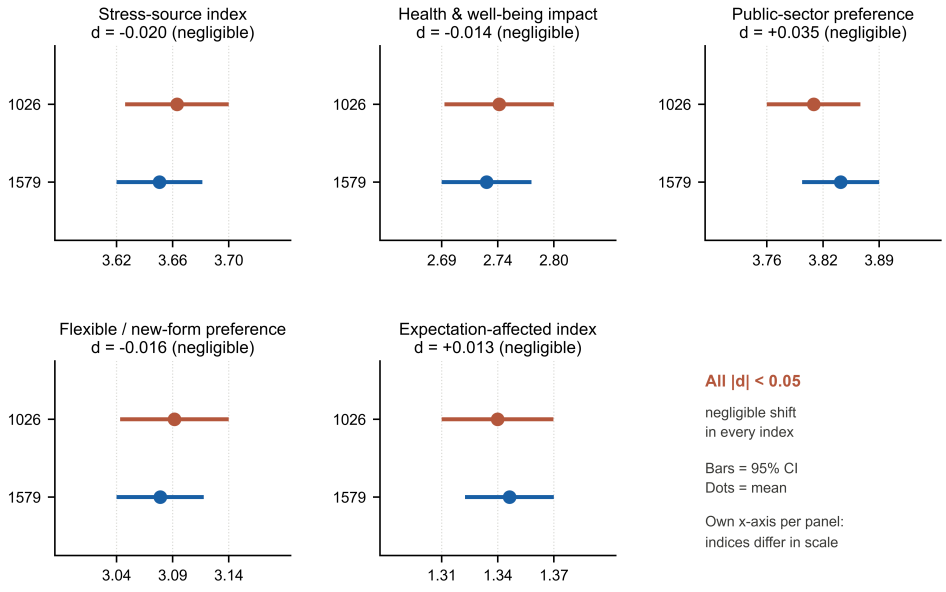


Fig. S4 Means and 95% confidence intervals of the five core indices in the main sample (dark red, upper row of each panel) and the pooled sample (blue, lower row). Each index occupies its own panel on its own horizontal scale, because the indices are measured on different ranges and a common axis would compress the differences of interest to a few pixels. Cohen's *d* and its magnitude label are given under each panel heading. At item level the largest displacement is $|d| = 0.052$ for the stress sources and 0.044 for the preferences. Reliability (main/pooled): Cronbach's α 0.839/0.845 for the nine stress sources, 0.676/0.682 for the eight preferences and 0.751/0.747 for the seven expectations; Kendall's *W* 0.085/0.093 and 0.160/0.176.

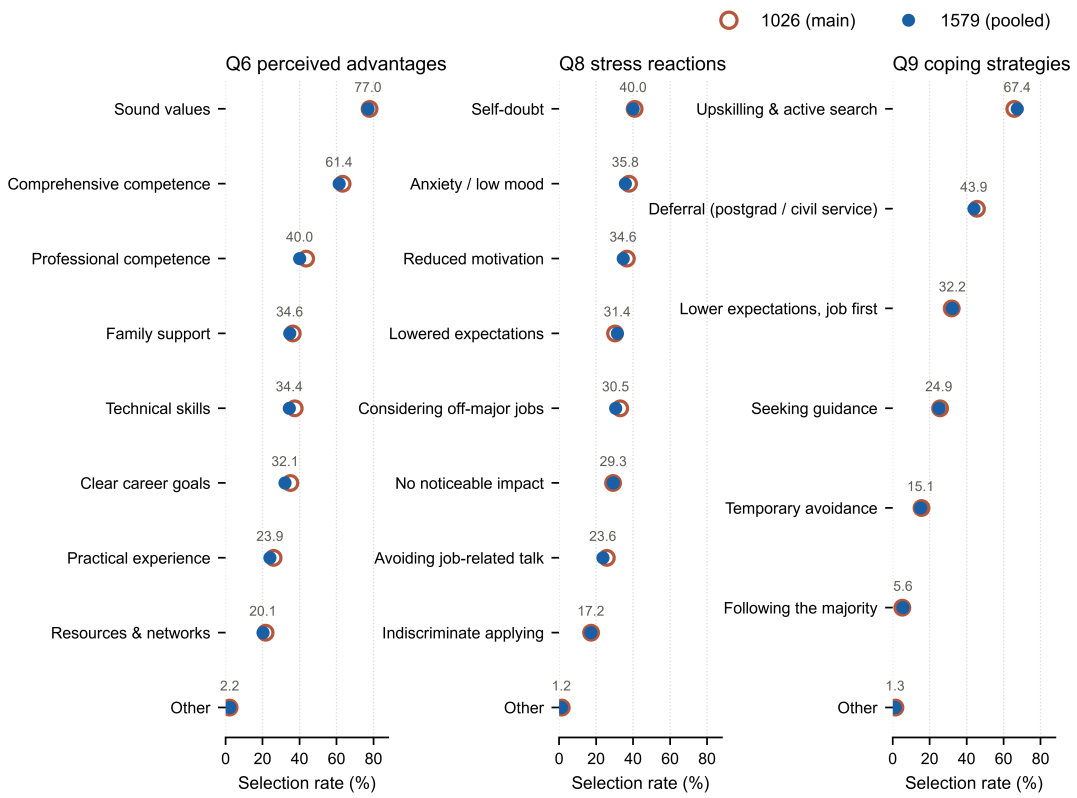


Fig. S5 Selection rates in the three multiple-response blocks — (a) self-assessed advantages (Q6), (b) reactions to stress (Q8) and (c) coping strategies (Q9) — for the main sample (open circles) and the pooled sample (filled circles). Rows are ordered by the pooled selection rate, the value of which is printed above each pair of markers. The three blocks share one percentage scale, so selection rates can be compared across blocks. The largest shift in selection rate is 3.44 percentage points.

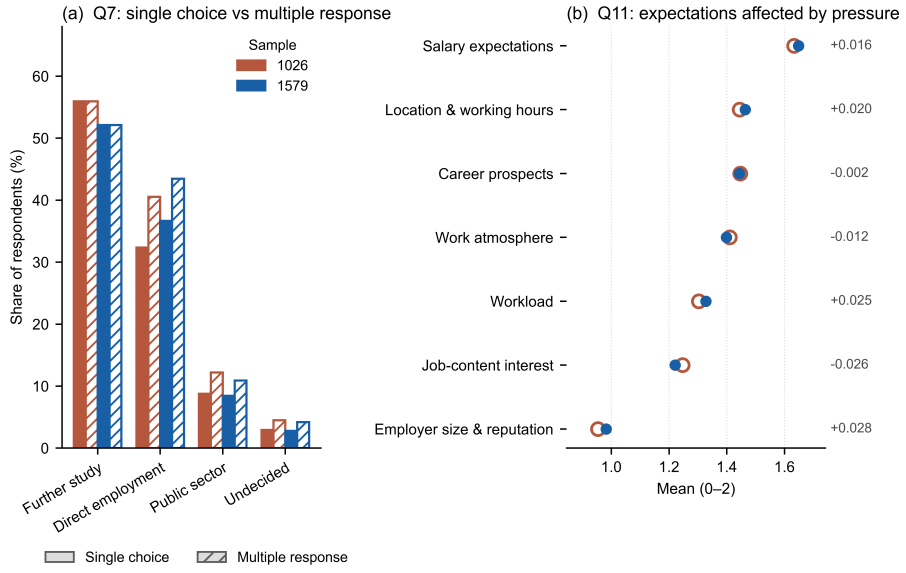


Fig. S6 (a) Preferred pathway after graduation (Q7) under the single-response frame (solid bars) and the multiple-response frame (hatched bars), for the main sample (dark red) and the pooled sample (blue); within each category the four bars are, from left to right, main sample single, pooled single, main sample multiple and pooled multiple. The gap between the solid and the hatched bar of the same colour is the ratio of acceptable to preferred mentions. (b) Mean scores of the seven expectation items (Q11, 0–2 scale) in the two samples (open circles, main sample; filled circles, pooled sample), with the difference in means on the right of each row. The ratio of acceptable to preferred mentions falls from 1.39× in the main sample to 1.29× in the pooled sample, and from 1.53× to 1.50× for undecided respondents.